

1. Write a `Function<String, Integer>` that returns the length of a given string.
2. Create a `Function<Integer, Integer>` that returns the square of a number.
3. Convert a `Function<String, String>` to return the string in uppercase.
4. Write a function that takes a string and returns its reversed version.
5. Create a function that converts INR to USD (using a fixed rate).
6. Create a `Function<String, Boolean>` that returns true if a string contains only digits.
7. Convert a full name "First Last" into initials using `Function<String, String>`.
8. Create a generic method:

java

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```
public static <T, R> R applyFunction(T input, Function<T, R> func)
```

and use it to convert strings to uppercase.

9. Parse a list of numeric strings to integers using `Function` and `Stream API`.
10. Combine two functions: multiply a number by 10, then add 5. Use `andThen()`.

◆ Advanced Applications

11. Use `Function<Employee, String>` to extract employee names from a list.
12. Use `Function<Product, Double>` to calculate price + 18% GST.
13. Chain functions to:
 - Trim a string
 - Convert to lowercase
 - Add "-done"
14. Convert a list of salaries to net values (after 10% tax).
15. Count the number of words in a sentence using `Function<String, Integer>`.
16. Write a function to count vowels in a string.
17. Capitalize the first letter of each word in a sentence.

18. Remove all non-alphabetic characters from a string.
 19. Remove all whitespaces from a string.
 20. Implement a `Function<List<String>, Integer>` that returns total character count.
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◆ List & Stream Based

21. Convert a list of names to initials using Function with Stream API.
 22. Apply 10% discount to a list of product prices.
 23. Convert a list of Employee to a list of salaries using `Function<Employee, Double>`.
 24. Sort a list of strings by length using Function and Comparator.
 25. Filter out empty/null strings using Function with `.filter()`.
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◆ Function Chaining

26. Chain 3 functions: Add 2 → Multiply by 3 → Append " Rs".
 27. Show difference between `compose()` and `andThen()` with an example.
 28. Trim, lowercase, and remove digits from a string using a function chain.
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◆ Advanced Use & Design

29. Build a transformation engine using `Map<String, Function<String, String>>`.
 30. Write a method that takes a `Function<T, R>` and a list, returning a transformed list.
 31. Mask an email address:
"sejal123@gmail.com" → "sej****@gmail.com"
 32. Convert a comma-separated string to a list of uppercase words.
 33. Generate usernames from full names: "Sejal Patankar" → "sejal.p".
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◆ Trick & Conceptual Applications

34. Can a `Function<T, R>` throw a checked exception? How to handle it?

35. Create a custom functional interface like `Function` that supports checked exceptions.
 36. How does `Function` support lazy evaluation in Stream API?
 37. Build a validation pipeline using Function chaining (e.g. for password checks).
 38. Format log messages using `Function<LogEntry, String>`.
 39. Dynamically select a sorting strategy using `Function`.
 40. Use a `Function<Object, String>` to serialize objects to JSON (e.g., using Jackson).
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◆ High-Level & Theoretical

41. Where have you used `Function` in your project? Explain use case.
 42. What are some disadvantages of chaining too many functions?
 43. When is `Function` better than traditional method calling?
 44. How does `Function` help write testable, modular code?
 45. Can `Function` improve performance in concurrent environments?
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◆ Bonus

46. Create a `Function<String, Integer>` to return ASCII sum of all characters.
47. Check if first character of a string is uppercase using `Function`.
48. Pass a `Function` as a method parameter and apply it.
49. Store named transformations (upper, reverse, trim) in `Map<String, Function<String, String>>`.
50. What is the difference between `Function<T,R>` and `UnaryOperator<T>`?